Classification of Marine Accident Investigation Reports Using Transformer-Based Language Models

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***Abstract*—[TODO: 150–250 words. Problem -¿ gap -¿ method -¿ dataset -¿ key quantitative result -¿ implication. Write this LAST.]**

***Index Terms*—maritime safety, accident reports, text classification, BERT, natural language processing**

I. Introduction

[TODO: Why marine incident reports matter; why manual classification does not scale; the gap; your contribution.]

Contributions of this paper:

- [TODO: Contribution 1 – e.g., a labeled dataset built from MAIB reports]
- [TODO: Contribution 2 – e.g., systematic comparison of models]
- [TODO: Contribution 3 – e.g., error analysis / practical insight]

II. Related Work

A. Text Mining in Maritime Safety

[TODO: Prior work on accident report analysis.]

B. Transformer Models for Domain-Specific Classification

BERT [1] [TODO: BERT and domain adaptation.]

III. Dataset

[TODO: Source (MAIB), collection, preprocessing, label definitions, class distribution.]

TABLE I
Dataset Statistics

Class	# Reports	%
Class A	0	0.0
Class B	0	0.0
Total	0	100.0

IV. Methodology

[TODO: Model architecture, input handling (long reports vs. 512-token limit), fine-tuning procedure.]

V. Experimental Setup

[TODO: Train/val/test split, baselines, hyperparameters, hardware, evaluation metrics (macro-F1 for imbalance).]

VI. Results and Discussion

TABLE II
Classification Performance

Model	Precision	Recall	Macro-F1
Baseline (TF-IDF + SVM)	–	–	–
BERT	–	–	–

As shown in Table II, [TODO: interpretation].

VII. Conclusion

[TODO: Summary, limitations, future work.]

Acknowledgment

[TODO: Optional.]

References

[1] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2019, pp. 4171–4186.